



An integrated analysis of sinkholes in Kadapa region, Andra Pradesh, India: Implication to pedology

M. Madakka^{a,*}, N. Jayaraju^b, J. Shirisha^c

^a Department of Biotechnology and Bioinformatics, Yogi Vemana University, Kadapa 516 005, A.P., India

^b Department of Geology, Yogi Vemana University, Kadapa 516 005, A.P., India

^c Department of Biotechnology and Bioinformatics, Yogi Vemana University, Kadapa 516 005, A.P., India

ARTICLE INFO

Keywords:

Agriculture soil
Sink holes
Heavy metals
Physicochemical properties
SEM
FTIR
XRD
Microbial population

ABSTRACT

Sinkholes are one of the common karst features of geographical formations, which come into existence due to the dissolution of underneath lime-rich formations and subsequent collapse of the ceiling of the holes. Since sinkholes are formed in agricultural lands, this study aimed to understand the impact of sinkhole formation on heavy metals and minerals by collecting samples from sinkholes from various localities of Kadapa District of the state of Andhra Pradesh in India. The heavy metals were determined by ICP-AES analysis and characterization of soils in sinkholes was carried out. The FTIR spectroscopic technique was utilized to identify the minerals present in sinkholes. The XRD and SEM techniques were used to assess clay mineral composition and particle size and morphology of soil samples, respectively. Following the available literature, the minerals were identified from the location or band position of peaks from the Infra Spectrum. The minerals found were Quartz, Cesium di alumino tri silicate hydrate, Natrolite (Cs), Sodium Rhodium Oxide, Cobalt Uranium, Cesium Manganese Fluoride Hydrate, Quartz alpha, alpha-Si O₂, Di copper (I) oxide, Erbium Indium Silver. Subsequently, we assessed the consequence of heavy metal pollution on soil microorganisms in sinkholes. The total population of microorganisms (bacteria, fungi, actinomycetes) was found to be decreased with increasing concentration of heavy metals. Additional analysis identified the bacteria resistant to Zinc, Manganese, lead (2 mM), copper (5 mM) metals. In sum, the findings from our integrated analysis of sinkholes provide an understanding of local pedology of the Kadapa region in India with implications for safety measures, agriculture land management and productivity.

1. Introduction

Agriculture plays a central role in the national economies of various countries across the globe including India. It is influenced by several factors, some of them are environmental in nature (e.g., climate and soil properties) and the others relate to anthropogenic activities (e.g. agriculture practices and irrigation). The soil is a key component of rural and urban environments, and land utilization, management, and monitoring in these areas are vital to soil quality. Globally, anthropogenic soil contamination is alarming, especially those related to heavy metal pollution of surface soils due to intense industrialization and urbanization, particularly in several developing countries [1–3]. Heavy metal contamination degrades soil quality, reduces crop yield and the quality of agricultural products, in turn posing major threats to flora and fauna and the existing ecosystem [4,5]. Soils are getting polluted with

accumulation of the toxic substances such as metals, trace elements, and other organic substances from domestic and industrial sectors [6], and such soil contaminants interfere with various key biochemical processes that alter the ecological settings [4,7].

Given that soil is a pivotal component of agriculture practice and success, the field of Agronomy as a branch of agricultural science deals with principles and practices of soil, water, and crop management. It helps to increase crop yield based on the criteria of homogeneous conditions such as rainfall, temperature, soil, topography, cropping and farming systems and water resources [8]. Thus, proper management of soil as a valuable resource is vital to sustain long-term agricultural productivity. Unfortunately, sinkholes are usually identified as a problem in agricultural fields. A sinkhole is a depression or hole in the ground caused by some form of the collapse of the surface layer, which can occur due to natural and/or anthropogenic processes. Sinkholes can

* Corresponding author.

E-mail address: madakka@gmail.com (M. Madakka).

<https://doi.org/10.1016/j.microc.2021.106588>

Received 22 October 2020; Received in revised form 26 June 2021; Accepted 28 June 2021

Available online 4 July 2021

0026-265X/© 2021 Elsevier B.V. All rights reserved.

be hazardous or even catastrophic, causing problems in agricultural areas and related activities [9].

Therefore, the focus of this study is on a rural area in Kadapa district of the state of Andhra Pradesh, India, where sinkholes have become problematic for local people [10]. The study area represents people of several villages of Chintakommadinne mandal (i.e., mandal refers to a local government area) and Vempalle mandal in Kadapa district, who have been affected by sinkholes. The formation of sinkholes with a diameter of 25 m at several places in the villages of Nayanoripalle, Goodavadlapalli, and Gindengivarepalle have become worrisome for the villagers. In addition to the sinkholes that evolved overtime, there have been newly occurring sinkholes in this region, especially those happening rapidly in the agricultural fields after heavy rainfalls. Erratic water movements in areas underlain by limestone may cause the land surface to become unstable, which can result in the sinkhole deepening or widening, and in some cases resulting in sudden surface collapse [11]. Such abrupt incidences and sudden surface collapses can be hazardous to human life, livestock, settlements, and agricultural lands; indeed, they have become alarming situations for the people living in this area.

Given attention to the issue of land subsidence discussed above, this field study aimed to conduct an integrated analysis of soil samples collected from sinkholes, particularly those found in the agricultural fields of selected villages.

2. Experimental section

2.1. Selection of sampling station

Soil samples were collected from different fields of the area of investigation, representing three villages as described above. In total, four sinkholes in agricultural fields were selected with different depths (9.15, 3.66, 4.575, 3.05 m).

2.2. Sampling procedure

Collection of soil samples was done using the auger method. Collected soil samples were placed in plastic bags, which were sealed, and labeled appropriately. Samples from each of the four sinkholes with different depths were collected randomly, and transferred to the laboratory for analysis without any outward contamination. The soil samples were dried thoroughly, and were sieved through a 2-mm sieve. The samples were kept intact in hygiene polythene bags for future analysis. The soil samples were collected from three villages of the Kadapa region of Andhra Pradesh: two samples (S1 and S2) from Nayanoripalle; one sample (S3) from T. Velamavaripalli; and one sample (S4) from Goodavadlapalle. The soil samples were collected in polyethylene bags using standard protocols; and, were analyzed to measure various physicochemical parameters by standard methods. Advanced analytical techniques were employed for the study as detailed below, and analysis of soil was carried out for examining various parameters like Total Organic Carbon, Total Nitrogen, Nitrate ($\text{NO}_3\text{-N}$) and silt, clay, sand, water holding capacity, electrical conductivity, and pH.

The heavy metal analysis was done by using ICP-AES (Inductively coupled plasma – Absorption emission spectroscopy). The X-ray diffraction (XRD) analysis was employed both quantitatively and qualitatively to examine the composition of the clay minerals in the samples. Fourier transform infrared spectroscopy (FTIR) analysis was carried out to identify the chemical structure of organic and inorganic molecules, which comprise covalent bonds. Soil microorganisms play a key role in the maintenance of soil fertility, fabric alteration and bio mineralization of organic matter, making nutrients available for plants. Every gram of a typical healthy soil is home to several thousand different species of bacteria. Although soils contain actinomycetes, bacteria, and fungi, their relative amounts depend on soil conditions. Soil microbial community and plant growth and yield are adversely affected by heavy metal contamination. In this context, this study also examined the impact of

selected metals on soil population of non-target microorganisms, following similar approach employed elsewhere [12].

2.3. Study site and location map

Kadapa (known as Cuddapah also) is a town in the Rayalaseema region of the south-central part of the state of Andhra Pradesh, India. It is the administrative headquarters for the Kadapa district (currently called as YSR district), which lies between latitude $79^{\circ}1'28.759''$ E and longitude $15^{\circ}5'3.875''$ N (Figs. 1 & 2).

2.4. Heavy metal analysis

Total metal concentrations were determined following a microwave-assisted acid extraction method [13]. 250 mg samples of sediment were transferred to a 15 ml polypropylene tubes of 10 ml of extractant mixture ($\text{HNO}_3 + \text{HF}$) [14]. Tubes were vortexed and irradiated for 30 sec using a microwave (640 W). The polypropylene tube caps were opened after irradiating the samples for 10 sec to let out gaseous products produced during microwaving (900–2400 MHz). Tubes were then closed and the microwave extraction process was continued for 20 sec for a total of 30 sec. Following extraction, supernatant was separated from the undissolved residues (silicates, etc.) by centrifugation for 4 min at 5000 rpm and then the clear supernatant was transferred to another pre-cleaned tube and diluted to the required volume for subsequent analysis by ICP-AES and high resolution continuum source Electro thermal atomic absorption (ETAAS) method at the BARC-National Center for Compositional Characterization of Materials (NCCCM), Hyderabad, Telangana. In the same manner, corresponding process blanks were also prepared without any sample materials. For the extraction process, three different aliquots of each sample were used. All the analytical measurements were run in triplicate for each sample solution. Results were presented as mean $\pm$ SD in Table 1.

The heavy metal analysis of concentrations of zinc (Zn), lead (Pb), nickel (Ni), manganese (Mn), cadmium (Cd), cobalt (Co), and chromium (Cr), was performed using a digestion method of acid dissolution through the technique of ICP-AES, representing calibration between 0.5 and 4 ppm. One gm of the soil sample was taken from each sinkhole for analysis of heavy metals by ICP-AES and high-resolution continuum source Electro thermal atomic absorption method (ETAAS) with environmental conditions of 48% humidity and 21.5°C temperature at the Bulk analysis section, BARC-NCCCM, Hyderabad.

2.5. Physicochemical analysis

For physicochemical analysis, the soils samples were scrutinized in the laboratory by using standard methods. The soil, sand, silt and clay content were determined by using mechanical analysis of Pipette method. Ten grams of soil sample suspended in 1 dm^3 of deionised water were passed through a set of American Society for Testing and Materials (A.S.T.M.) sieves ($1000 - 32\text{ }\mu\text{m}$). The transmitted suspension was then poured into a 1 dm^3 graduated cylinder and allowed to settle. Successive aliquots of 10 cm^3 were extracted from the suspension at the time intervals calculated for a specific particle diameter (according to Stokes' law), dried and weighed. pH was determined by using a pH meter (Elico LI617); electrical conductivity was determined by using a conductivity meter (Elico CM 180); and, water holding capacity was determined. The micronutrients analysis of manganese (Mn), zinc (Zn), copper (Cu) and iron (Fe) was performed using Atomic Absorption Spectrophotometer (AAS). Organic carbon was determined using the Walkey and Black method [15]. 1.0 g of sieved soil was weighed into a 500 ml Erlenmeyer flask. 10 ml of 0.1 M potassium dichromate was added. The soil was dispersed by swirling and then 20 ml of concentrated sulphuric acid was immediately added to wet the sample. 200 ml of water was added with 4–5 drops of Ferroin indicator. The resulting mixture was titrated using 0.5 M potassium permanganate organic Carbon, nitrite, and nitrate; and,

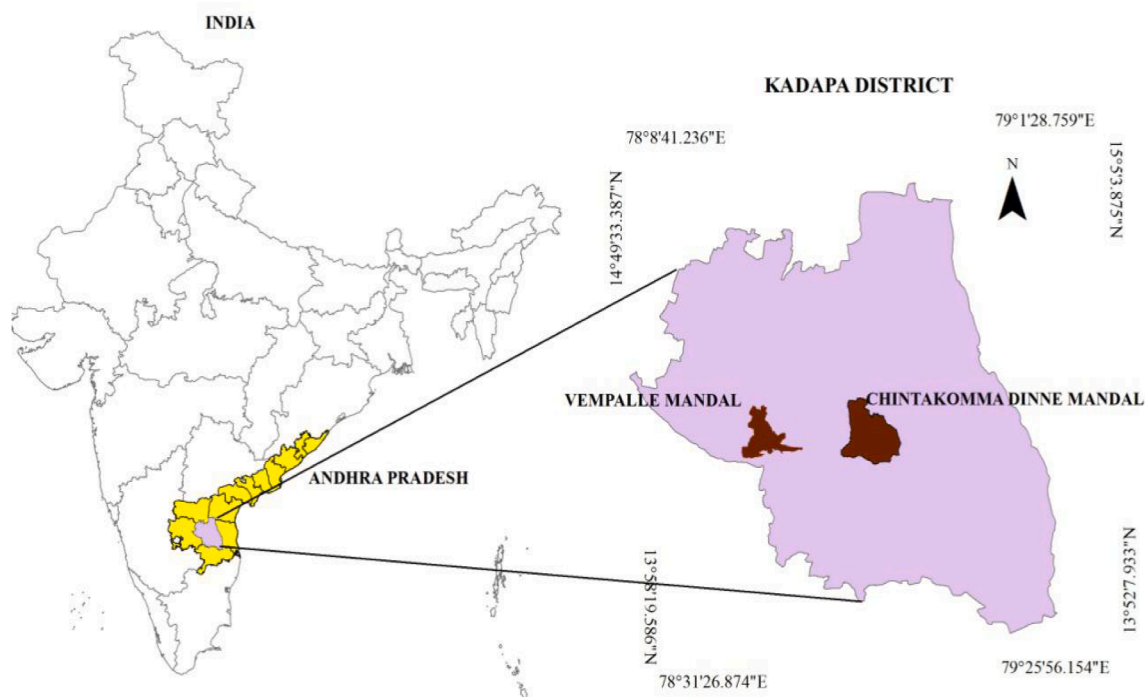


Fig. 1. Study area of sink holes.

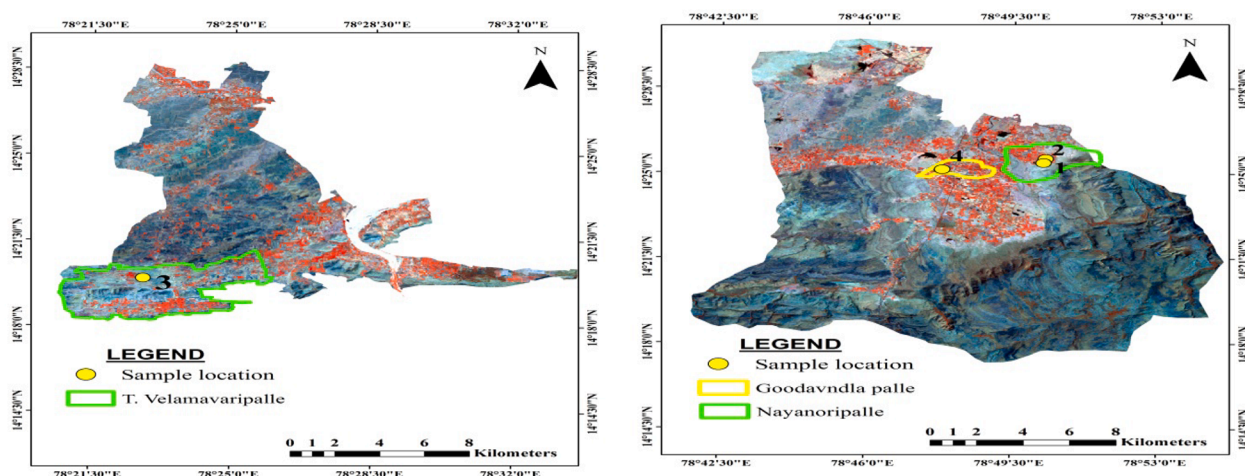


Fig. 2. Satellite images around the sinkholes.

Table 1

Heavy metal analysis in four sites of sink holes.

Samples	Cd	Co	Cr	Mn	Ni	Pb	Zn
Soil 1	<1	6.1 ± 0.4	40 ± 3	235 ± 20	18 ± 2	2.1 ± 0.2	26 ± 2
Soil 2	<1	9.0 ± 0.6	40 ± 4	423 ± 30	25 ± 2	3.1 ± 0.2	23 ± 2
Soil 3	<1	12.0 ± 0.9	87 ± 5	1500 ± 0.01	39 ± 3	1.9 ± 0.1	17 ± 1
Soil 4	<1	6.9 ± 0.4	35 ± 3	264 ± 22	16 ± 1	2.9 ± 0.2	12 ± 1

Units: All samples are represented in ppm (mg/kg).

calculation of organic carbon was done as organic carbon (%) = $10(B-T)/B \times 0.003 \times 100/\text{wt. of soil (g)}$. Total nitrogen was determined following the method of the Alkaline permanganate method by Subbaiah and Asija [16], using KELPLUS Assembly. Available N (kg ha^{-1})

was calculated using the formula: $(S-B) \times 0.02 \times 0.014 \times 2.24 \times 10^6 / \text{weight of soil (g)}$, and available nitrogen = $R \times 12.7$. Micronutrients like Zn, Cu, Mn and Fe available in soils were determined by the method developed by Lindsay and Novell8 using DTPA (Diethyl Triamine Penta Acetic Acid), which was found useful for separating soils into deficient and nondeficient categories for Zn, Cu, Mn, and Fe by using atomic absorption spectrophotometer.

2.6. Sample preparation for FTIR analysis

Fourier transform infrared (FTIR) spectral measurements were recorded (Perkin Elmer, Spectrum Two Model, UK) to confirm the content of organic matter present in the soil sample. Approximately 2 mg of sample was mixed with 200 mg of spectroscopic grade KBr at a ratio of 1:20 using a mortar and pestle. The necessary amount of KBr powder was dried at 120 °C for six hours in a hot air oven before mixing, since the broad spectral peak that resulted due to free OH will seriously

affect the interpretation of the bound hydroxyls which are associated with any of the minerals. The dry mineral sample was weighed in a micro balance and placed in a clean agate mortar along with the proper amount of dry KBr to prepare a sample pellet. A pellet of 1 mm in thickness and 13 mm in diameter was prepared under a pressure of 600 Kg/cm² using the hydraulic press. This method is adopted for the preparation of every pellet [17]. The prepared pellet was then placed in the sample holder and introduced into the infrared beam for spectral analysis in the range of 4000–450 cm⁻¹. For each site, five to six samples were collected and for each sample, five to six pellet specimens were prepared and examined.

2.7. Sample preparation for XRD

The widely used method of mounting sample for X-ray analysis was performed for preparation of oriented sample on the microscopic glass slide. A clay suspension was made properly and pipetted out onto the glass slide. The XRD patterns of the soil samples were measured using the CuK_{α1} (1.5406 Å) rays (Power: 35 V × 15 mA) with a step size of 0.02 Å⁰ (RIGAKU; Miniflex –600).

2.8. Sample preparation for SEM analysis

Particle size and morphology of the soil samples were investigated by SEM-EDS analyses, and the morphological characters were obtained from SEM, supported by Energy Dispersive X-ray (EDS) micro analysis device. SEM analysis was carried out using FESEM MERLINE COMPAC S.NO.6027.CARL ZEISS electron microscope equipped with energy dispersive micro analysis system at the School of Chemistry, University of Hyderabad, Hyderabad, Telangana. The SEM photographs and corresponding EDS spectra were taken for all samples. Typical microphotograph and spectrum of soil samples are shown in Figs. S1 to S4.

2.9. Effect of selected metals on soil population of non-target microorganisms

2.9.1. The population of Bacteria

The effects of different concentrations of selected heavy metals viz., Zinc, Nickel, Lead, Manganese on bacterial populations from four sinkhole samples representing different agricultural soil samples, in triplicate, were determined. Aliquots (0.05 ml) from stock solutions of the pesticides were applied to two-gram portions of soil contained in test tubes (15 X 150 mm). The final concentrations of each metal 1 mM, 2 mM, 3 mM, 4 mM, 5 mM were added to the soil samples. The soil samples receiving only distilled water were then homogenized to distribute the metal. However, enough distilled water was added to maintain at 60% water holding capacity (WHC) and incubated at room temperature (28 ± 4 °C). Seven days after incubation, duplicates of each treatment were taken for estimation of the bacterial population. Aliquots were prepared (10⁻⁴ to 10⁻⁷) from treated and untreated soil samples by serial dilution plate method on nutrient agar medium and subsequently incubated for 24 h in an incubator at 30 °C [18]. Bacterial colonies were grown on nutrient agar medium after incubation, and were counted by Quebec colony counter. Bacterial colonies were enumerated, expressed as number of colonies formed per gram of soil (dry weight basis) [19].

2.9.2. The population of fungi

The impact of different concentrations of the selected metals on the fungal population in four different agricultural soil samples, in triplicate, was determined, following the same procedure adopted as in the case of the bacterial population. Soil plate method was used to assess fungal propagules developing on Potato Dextrose agar medium; subsequently, it was incubated for five days at 25 °C [20,21,18].

2.9.3. The population of actinomycetes

The population of actinomycetes was estimated by using Kenknights

agar medium and subsequently incubated for three days in the dark at 30 °C [22].

3. Results and discussions

3.1. Heavy metal analysis

The heavy metals data generated from soils of all the sites are shown in Table 1. All the metals studied were detected at all the sites. For estimation of heavy metal (Cd, Co, Cr, Mg, Ni, Pb, Zn) concentrations, 1 g of dried soil was digested with concentrated HNO₃, H₂SO₄ and H₂O₂ (2: 6: 6) [23]. The blanks were run in a set, and samples were analyzed with an Inductive Coupled Plasma Analyzer at Sophisticated Instrumentation at National Center for Compositional Characterization of Materials (NCCCM), BARC, Hyderabad. The patterns of concentrations of the heavy metals found from different sites are as follows. The Site 1 sample exhibited the higher concentration of metals in the following order: Mn > Cr > Zn > Ni > Co > Pb > Cd; the Site 2 sample showed the higher concentration of metals in the following order: Mn > Cr > Ni > Zn > Co > Pb > Cd; the Site 3 sample showed the higher concentration of metals in the following order: Mn > Cr > Ni > Zn > Co > Pb > Cd; and the Site 4 sample exhibited the higher concentration of metals in the following order: Mn > Cr > Ni > Zn > Co > Pb > Cd. The heavy metals including manganese from all the sites were found to be at higher concentration levels, which were expressed in terms of ppm or mg/kg.

3.2. Physico-chemical properties

The physicochemical parameters of the effluents investigated are presented in Table 2.

3.2.1. Pipette method

The fractions of different horizons of the proportions of sand, silt and clay particles in the soil are shown in Table 2. Sand, silt, and clay particles refer to different grain size fractions of the sample content: 0.06–2

Table 2
Physicochemical parameters.

Physical Properties	Methods	Site 1	Site 2	Site 3	Site 4
		Black soil		Red soil	
Silt (%)	Pipette analysis	16.253	22.459	15.639	13.272
Clay (%)	Pipette analysis	2.616	3.291	3.251	1.718
Sand (%)	Pipette analysis	81.12	74.25	81.1	84.99
pH	pH meter (Elico LI617)	7.84	8.26	8.12	7.76
Water holding capacity (ml/g)		0.9	1.1	0.8	0.7
Electrical conductivity (EC) (dSm ⁻¹)	Conductivity meter (Elico CM 180)	0.17	0.22	0.20	0.09
Chemical properties					
Organic carbon (%)	Walkley and Black, method(1934)	0.43	0.75	0.16	0.001
Available nitrogen(kg/mg)	Alkaline permanganate method,(1956), using KELPLUS assembly	63	88	13	13
Micronutrients (concentrations in ppm)					
Copper (DTPA extractable)	Atomic Absorption Spectroscopy(4141)	0.548	0.726	0.786	0.488
Iron (DTPA extractable)		3.750	3.512	4.778	5.254
Manganese (DTPA extractable)		3.254	3.698	3.550	4.328
Zinc (DTPA extractable)		0.596	0.284	0.612	0.644

mm, 0.002–0.006 mm and < 0.002 mm [24].

3.2.2. pH

The pH of the soil is one of the most important physicochemical parameters. It affects mineral nutrient soil quality and microbial activity. The pH values of the soil samples in all the sites ranged from 7.76 to 8.26. The pH of soil samples found was basic in almost all the samples. Some of the soil properties are vital in soil processes including transportation of heavy metals.

3.2.3. Water holding capacity

The water holding capacity of soils at the sites ranged from 0.7 to 1.1 (ml/g), and it was higher in site 2. Similar findings were also reported by Sathyavathi and Suryanarayan Reddy [25] regarding the soils of Sivagiri micro-water shed in Chittoor district of Andhra Pradesh and the Telangana region.

3.2.4. Electrical conductivity

The Electrical Conductivity study of soil samples showed variation in conductivity values, which ranged from 0.09 dSm⁻¹ to 0.22 dSm⁻¹, suggesting that it was normal soil. The site 2 exhibited high electrical conductivity compared to the sites 1, 3 and 4. [26,27].

3.2.5. Organic carbon

Percentage of Organic carbon varied from 0.001 to 0.75 different values as shown in Table 2. As can be seen, the sample from site 4 was found to have high organic carbon compared to the samples from the other three sites. This observation is in concordance with the findings reported by Tsui et al. [28].

3.2.6. Available nitrogen

The available nitrogen content in all the soil samples ranged from 13 to 18 mg/kg. The sites 3 and 4 red soils have equal of nitrogen content and site 2 sample has high nitrogen content as compared to sample have high nitrogen content as compared to sites 1, 3 and 4.

3.2.7. Micronutrients

Micronutrients are analyzed by the atomic absorption spectroscopy, these results of four sites shown on Table 1, The concentrations of Cu range between 0.488 and 0.786 ppm indicating enrichment of Cu in the soils (optimum range for Cu is 0.2 to 2.5 ppm). Here all sites represented the sufficient. Iron content in analyzed samples was high 5.254 ppm and 3.512 ppm as the normal range for Fe is 2.5 to 4.5 ppm (Table 2). Site 3, and site 4 samples represent sufficient symptoms and site1, site2 samples represent the deficiency. The soil samples from the study area show manganese content from 3.254 to 4.328 ppm which is highly enriched in Mn (normal soils Mn range 1.02 – 2.5 ppm). Here the four sites of Manganese represent the sufficient. Soils enriched with Zinc range between 0.284 and 0.644 ppm as in normal soil, it ranges from 0.51 to 1.21 ppm. Site 3, and site 4 has sufficient value of Zinc and site 1, site 2 has the deficient value of Zinc.

3.3. FTIR analysis

FTIR analysis of soil samples (S1, S2, S3, and S4) was carried out based on earlier literature [29] and the results are represented (Fig. 3). From the results, it was observed that the peaks at about 3696 and 3620 cm⁻¹ (S1, S2, S3, and S4) are corresponding to OH groups (alcohols, phenols, organic acids) of organic matter present in the sediments. These absorption bands have also confirmed the presence of hydroxyl vibrations of clay minerals such as silanols (Si-OH) and aluminous (Al-OH) which can accumulate at the surface of the clay. These functional groups can be involved in the complexation of metal cations.

An intense vibrational band found at around 3435 cm⁻¹ is resulted from the elongation of molecular water that is absorbed by the inorganic material present in the sediment or by the KBr pellet. Twin peaks at 2925

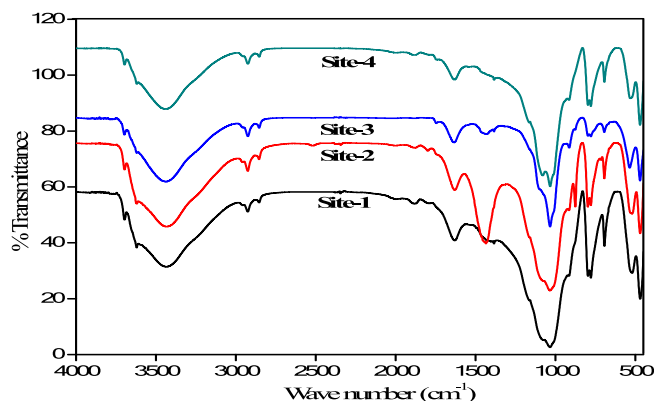


Fig. 3. FTIR analysis of four sites of sink holes.

and 2854 cm⁻¹ are attributed to C-H vibrations of aliphatic methyl and methylene groups. A very small intense peak at 2519 cm⁻¹ is found for S2 corresponds to typical vibrations of calcite. A group of very less intense three peaks centered at 1995, 1890 and 1793 cm⁻¹ assigned to quartz overtones. A vibrational band around 1630 cm⁻¹ (for all samples) corresponds to stretching absorptions of olefinic and aromatic C = C and C = O group of carboxylic acids. A shoulder peak at about 1430 cm⁻¹ (S1, S3, and S4) is attributed to carbonates superimposed on bending vibrations of C-H bonds of methyl groups. But this peak is more intense in the case of S2.

Further two high intense characteristic peaks observed at 1088 and 1034 cm⁻¹ confirmed the presence of phosphates in the sediment samples. A little intense deformation band is observed at around 907 cm⁻¹ typically assigned to Al-OH (all samples). In S2, a sharp less intense vibrational band is appeared at 873 cm⁻¹ also confirms the presence of calcite in the sample. The more intense characteristic peak in-between 803 and 775 cm⁻¹ is appeared due to the associated minerals or crystal defects of quartz present in the sediment. The vibrational band of C-S stretching was found at 684 cm⁻¹ for all samples. The two peaks centered at around 530 and 470 cm⁻¹ are corresponding to Si-O-Al^{IV} and Si-O-Mg^{IV}.

3.4. XRD analysis

The powder XRD pattern of agricultural soil samples of site 1, 2, 3 and 4 were shown (Fig. 4). It revealed all of the reflection in the profile are well indexed to a single phase hexagonal structure belonging to P 3221 space group (no: 154, z = 3) (ICDD data base no 01-075-6025), the average crystalline size of sample is calculated by Scherer's equation and is found to be around 40–80 nm (Table 3).

The XRD results of soil samples obtained are given (Table 3 and 4). The XRD diffractograms are presented (Fig. 4). The XRD pattern of selected four soil samples (S1, S2, S3, and S4) was analyzed. Key XRD patterns of soil samples in different locations are shown (Fig. 4). The d-spacing values of site 1 is obtained from diffractograms presented in Fig. 1 are 4.2539, 3.3406, 2.4564, 2.2376, 2.12756, 1.81731, 1.5054 etc. It was found to have characteristics value of the essential minerals such as Quartz, Cesium dialuminotrisilicate hydrate, Natrolite (Cs) etc, The d-spacing values of site 2 soil sample is obtained from diffractograms presented in Fig. 4 are 4.25034, 3.3478, 2.45705, 2.23875, 2.12926, 1.81937, 1.5053, 1.38244 etc., In this sample, characteristics value of the minerals such as Quartz, Sodium Rhodium Oxide etc., were recorded. Site 3 d-spacing values is obtained from diffractograms presented in Fig. 4 are 4.2594, 3.3382, 2.4588, 2.2327, 2.12901, 1.81876, 1.4968 etc. In site 1, it was found to have characteristics value of the minerals such as Quartz, Cobalt Uranium, Cesium Manganese Fluoride Hydrate etc., and the d-spacing values of site 4 soil sample is obtained from diffractograms presented in Fig. 4 are 4.2596, 3.3597, 2.46053, 2.23889, 2.12954, 1.82031, 1.5061, 1.38287 etc. It was found to have

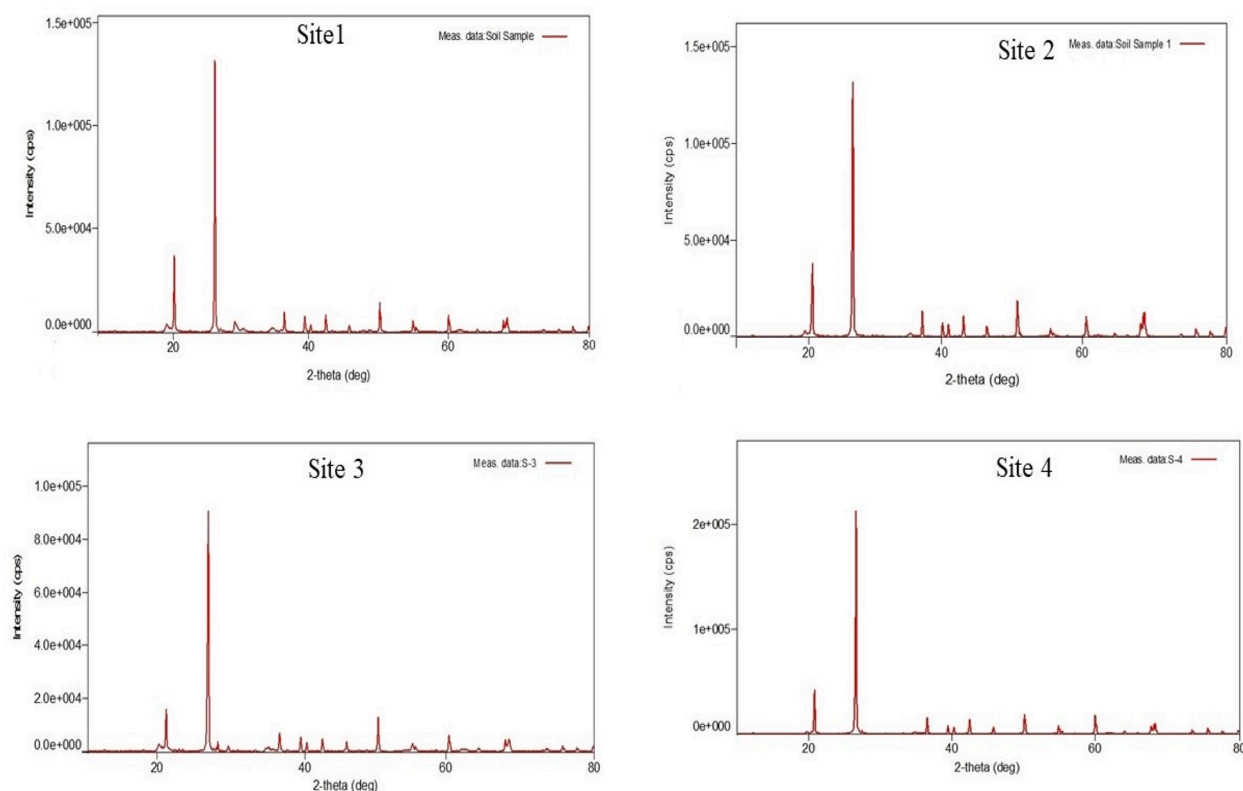


Fig. 4. XRD patterns of soils from sites 1, 2, 3, and 4.

Table 3

Shows the unit cell parameters and crystallite sizes calculated by Debye scherrer method.

S.No	Lattice parameter ($\text{\AA}$)		Crystallite size (nm)
	a = b	c	
1.	4.914 = 4.914	5.4031	66 nm
2.	4.9153 = 4.9153	5.4083	50 nm
3.	4.9169 = 4.9169	5.4055	39 nm
4.	4.9292 = 4.9292	5.4363	78 nm

Table 4

General Characteristics and minerals observed by XRD analysis in sinkholes Soils.

Sample	Sites of sample collection	Depth (ft)	Main Minerals Observed
Site 1	Nayanoripalle	30	Quartz, Cesium di alumina tri silicate hydrate, Natrolite (Cs)
Site 2	Nayanoripalle	12	Quartz, Sodium Rhodium Oxide
Site 3	Giddengivaripalli	15	Quartz, Cobalt Uranium, Cesium Manganese Fluoride Hydrate
Site 4	Goodavandlapalli	10	Quartz, Dicopper (I) oxide, Erbium Indium Silver.

characteristics value of the minerals such as Quartz, Di copper (I) oxide, Erbium, Indium, Silver etc.

3.5. SEM analysis

Out of the XRD analysis of the soil sample of sink holes from four different stations such as sites 1, 2, 3 and 4 have to characterize soil samples by Scanning Electron microscopy (SEM). As the particle size is $<0.2 \mu\text{m}$ cannot be resolved under the ordinary microscope and hence

cannot be seen. The arrangement of particle size average of the obtained liquid phase was investigated by SEM. Figs. S1 to S4 showed the microscopy of typical clay particles size from different soils. There are different spacing (different clay minerals) in the same crystal, confirming the presence of interstratified clay. SEM photograph clearly indicates the presence of quartz, cesium, natrolite, sodium rhodium oxide, cobalt uranium, dicopper oxide, Erbium Indium Silver. These results also demonstrate the heavy metal content in soil samples. Sites 1, 2, 3 and 4 were chemically analyzed for different parameters. This analysis clearly reveals that some of the parameter's value has been found to be higher. This analysis also supports the presence of detected minerals. The utility of the SEM of the surface texture to decipher different sedimentary environment has now been very well established. The careful study and investigation of sink holes soil samples from four stations indicated that the sediment indicates that the sediment sample contains quartz, cesium, natrolite, sodium rhodium oxide, cobalt uranium, dicopper oxide, Erbium Indium Silver.

3.6. Effect of selected heavy metals on the population of microorganisms in sinkhole samples

3.6.1. Bacterial population

The effect of Zinc, Nickle, Lead, Manganese on bacterial population was studied. Data on the effects of these heavy metals on bacterial populations are summarized in Figs. S5 to S8. Bacterial populations were significantly higher in Site 1 soil treated with lead in the medium. This increase in population was up to 4 mM at than in the untreated control, after 7 days of incubation (Fig. S5). Bacterial population in Site 4 was enhanced with increasing concentrations of four heavy metals especially in lead in the medium. As seen in the case of site 1 and site 4 soil the lead individually stimulates to the bacterial populations, at all tested concentrations (up to 4 mM) as reflected by the higher count of bacterial pollution in heavy metal treated media and control (without heavy metal) (Figs. S5 & S8). Bacterial populations were significantly higher in

Site 2 soil treated with lead and nickel in the medium. This increase in population was up to 5 mM at than in the untreated control, after 7 days of incubation (Fig. S6). Bacterial population in Site 3 was enhanced with increasing concentrations of four heavy metals especially in lead and nickel in the medium. As seen in the case of site 2 and site 3 soil the lead and nickel individually stimulate to the bacterial populations, at all tested concentrations (up to 5 mM) as reflected by the higher count of bacterial pollution in heavy metal treated media and control (without heavy metal) (Figs. S6 & S7). The capability of bacteria resistance against different heavy metals may offer a beneficial tool for the simultaneous monitoring of many contaminants and pollutants in the environment. It is noticeably evident that domestic and industrial wastes toxic effects are the reason why bacteria developed resistance mechanisms along with the hazard of human health and environment. Therefore, the study is very useful to suggest that the possible impact of metal contaminated locations in human life may be greater than the direct consequence of the pollution. Heavy metals contaminate the environment by gathering in the food chain and remains in nature. Each heavy metal has its sole toxicity. Copper and Zinc can improve microbial growth at low concentrations but suppresses growth at elevated concentrations [30]. To endure under metal-stressed circumstances, bacteria have evolved up to several types of adjustment to stand with the uptake of heavy metal ions. These mechanisms include the efflux of metal ions outside the cell, bioaccumulation of the metal ions inside the cell, and the decreasing concentration of the heavy metal ions. While the acute effect of heavy metals on the microbial community appears to lead to a subsequent shift in the community toward a more metal tolerant or metal-resistant population [31,32], in chronically contaminated sites natural selection should have resulted in a predominantly metal-tolerant community [33,34]. Many studies have focused on the effects of heavy metals on bacterial community structure [35–38].

3.6.2. Fungal Populations

Fungal populations in four sites decreased with increasing concentrations of all the heavy metals. In site 1 all fungi showed the tendency to increase in fungal growth with manganese in the medium (Fig. S5). In site 2 all fungi showed the tendency to increase in growth supplemented with Nickel, lead, and manganese in the medium (Fig. S6). In site 3 all fungi showed the tendency to increase in fungal growth with lead and Nickel in the medium (Fig. S7). In site 4 all fungi showed the tendency to increase in fungal growth with zinc in the medium (Fig. S8).

In the present study, considerable information was obtained regarding interaction effects between heavy metals towards fungal populations in soils. Synergistic/additive interactions between heavy metals in combinations at lowest concentration were predominant towards fungal population in soils whereas antagonistic response mostly figured in soils with heavy metals at the highest level in the present study. The Effect of Zinc, Nickel, Lead, and Manganese on fungal population was studied. The Data obtained from these experiments were furnished in Figs. S5 to S8. Soil pH greatly influences the solubility, available and toxicity of metal elements in soil [39]. Clayey – loamy agricultural soil was considered in this study. This partly shows the low toxicity. The clay mineral montmorillonite protects bacteria and fungi against inhibition by metal species [40]. Fungi were known to accumulate the significant amount of cadmium, for example, uptake concentrations of 6.46 mg g^{-1} by *Aspergillus niger* and 16.25 mg g^{-1} by *Trichoderma viride* have been reported earlier [41]. *Pisolithus tinctorius* presented maximal uptake of 600 mg kg^{-1} dry weight at 10 mg l^{-1} cadmium concentration [42]. The presence of multi-metals imposed a greater toxicity to fungi growth than that imposed by single metals. *Aspergillus niger* that had been adapted to tolerate Mo concentration up to 5000 ppm could not survive in a mixture which only contained 500 ppm of Mo. This suggests that a mixture of metals acts in synergy to impose a greater toxic response to the adaptation behaviour of fungi in comparison to single metals alone. Resistant fungal strains belonging to different genus have been found frequently in heavy metal

contaminated environments [43,44].

3.6.3. Actinomycetes

The effect of Zinc, Nickel, Lead, and Manganese on actinomycetes was studied. Data on the effects of these heavy metals on were tested for their effects on actinomycetes in four sites. The data obtained from these experiments were furnished in Figs. S5 to S8. In site 1, all actinomycetes showed the tendency to increase their growth with zinc in the medium (Fig. S5). In site 2, actinomycetes showed the tendency to increase in growth supplemented with lead in the medium (Fig. S6). In site 3, actinomycetes showed the tendency to increase their growth with Nickel in the medium (Fig. S7). In site 4, actinomycetes showed the tendency to increase their growth with manganese in the medium (Fig. S8).

Wang et al. [45] investigated the combined effect of cadmium (Cd) and butachlor on microbial activity. It is noted that the addition of high concentration of butachlor applied in combination with Cd significantly affected the diversity of the microbial population. A wide variety of microorganisms, for example, bacteria, yeast, algae, protozoa, and fungi, have developed the capabilities to protect themselves from heavy metal toxicity by various mechanisms such as uptake, oxidation, adsorption, methylation, and reduction [46]. It was also reported that some bacteria can use mechanisms of tolerance and detoxification of heavy metals and still produce chelating agents that bound metals and reduce their toxicity [47]. Actinobacteria constitute a morphologically diverse group of Gram-positive bacteria, with high metabolic versatility [48]. As reported earlier, actinomycetes are the good bioremediation [49–53]. Their metabolic diversity, particular growth characteristics, mycelia form, and relatively rapid ability to colonize selective substrates make them well-suited for use as agents for bioremediation of inorganic and organic compounds [51,54–57].

4. Conclusion

In the present research, four agricultural fields allowed us to carry out integrated analysis. The heavy metal (Zn, Pb, Ni, Mn, Cd, Co, & Cr) concentrations were found in all sites in the following order $\text{Mn} > \text{Cr} > \text{Ni} > \text{Zi} > \text{Co} > \text{Pb} > \text{Cd}$. In all the sites, Mn was found to be in higher concentrations. The study of physico-chemical parameters is necessary for agricultural chemists for crop production and land management. In all agriculture sites, silt range from 13.272% to 22.459%, clay from 1.718% to 3.251% and sand from 74.25 to 84.99%, respectively. Higher percentage of sand was identified in the agriculture soils. The pH of soil samples was found in basic nature in almost all the samples. The water holding capacity is higher in the site 2 as compared to all three sites. The Electrical Conductivity (Ec) in site 2 recorded higher values as compared to the sites 1, 3 and 4. From the results of FTIR, it was observed that the peaks at 3696 and 3620 cm^{-1} (S1, S2, S3, and S4) are corresponding to OH groups (alcohols, phenols, organic acids) of organic matter present in the sediments. The results of XRD analysis revealed that the vital minerals such as Quartz, Cesium di alumino trisilicate hydrate, Natrolite (Cs), cobalt uranium, dicopper oxide, Erbium Indium Silver etc were found in the study samples. From the results of SEM EDS analysis, it was inferred that the SEM photograph clearly indicates the presence of quartz, cesium, natrolite, sodium rhodium oxide, cobalt uranium, dicopper oxide, Erbium Indium Silver. These results also demonstrate the higher heavy metal content in soil samples. Samples from sites 1, 2, 3 and 4 were chemically analyzed for different parameters. The preliminary results showed that agricultural soil bacterial population might have decreased by the effect of high concentration of Ni and Pb. Similar studies, using soil of different level of metal effect and control should be carried out to draw a long-term impact of heavy metal pollution on soil microbial community to explore the possible metal-microbe interaction and their possible impact on soil health. So this observation was confirmed in this experimental of agricultural fields of sites 1, 2, 3 and 4 soil samples has decreased bacterial population, where the concentration of Mn and Zn (1–3 mM) and Pb and Ni (1–5 mM).

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.microc.2021.106588>.

References

- [1] F. Mireles, J.I. Davila, J.L. Pinedo, E. Reyes, R.J. Speakman, M.D. Glascock, Assessing urban soil pollution in the cities of Zacatecas and Guadalupe, Mexico by instrumental neutron activation analysis, *Microchem. J.* 103 (2012) 158–164.
- [2] B. Wei, L. Yang, A review of heavy metal contaminations in urban soils, urban road dusts and agricultural soils from China, *Microchem. J.* 94 (2) (2010) 99–107.
- [3] G. Yaylalı-Abanuz, Heavy metal contamination of surface soil around Gebze industrial area, Turkey, *Microchem. J.* 99 (1) (2011) 82–92.
- [4] B. Hu, J. Zhou, L. Liu, W. Meng, Z. Wang, Assessment of Heavy Metal pollution and Potential Ecological Risk in Soils of Tianjin Sewage Irrigation Region, North China, *J. Environ. Anal. Toxicol.* 7 (2017) 425.
- [5] P.C. Nagajyoti, K.D. Lee, T.V.M. Sreekanth, Heavy metals, occurrence and toxicity for plants: a review, *Environ. Chem. Lett.* 8 (3) (2010) 199–216.
- [6] M.S. Rakesh Sharma, N.S. Raju, Correlation of Heavy Metal contamination with Soil properties of industrial areas of Mysore, Karnataka, India by Cluster analysis, *Int. Res. J. Environ. Sci.* 2 (2013) 22–27.
- [7] B. Wang, H.T. Meng, Z. Zhang, The content features and potential ecological risk assessment of soil heavy metals in Tianjin suburban farmland, *Environ. Study Monitoring.* 4 (2010) 11–15.
- [8] P. Sahu, R.D. Lokhande, An Investigation of Sinkhole Subsidence and its preventive Measures in Underground Coal Mining, *Procedia Earth and Planetary Sci.* 11 (2015) 63–75.
- [9] M. Delle Rose, A. Federico, M. Parise, Sinkhole genesis and evolution in Apulia, and their interrelations with the anthropogenic environment, *Natural Hazards Earth System Sciences.* 4 (5/6) (2004) 747–755.
- [10] M. Prasad, M.R. Reddy, V. Sunita, Bedrock Structural Controls on the Occurrence of Sinkholes: A Case Study from Chintakommadinne Area, Part of Cuddapah Basin, South India, *J. Ind. Geophys. Union* 21 (2) (2017) 124–139.
- [11] A.A. Voorhies, B.A. Biddanda, S.T. Kendall, S. Jain, D.N. Marcus, S.C. Nold, et al., Cyanobacterial life at low O₂: community genomics and function reveal metabolic versatility and extremely low diversity in a Great Lakes sinkhole mat, *Geobiology* 10 (3) (2012) 250–267.
- [12] Z. Wang, X. Zeng, M. Geng, C. Chen, J. Cai, X. Yu, Y. Hou, H. Zhang, Health Risks of heavy metals uptake by crops grown in a Sewage Irrigation Area in China, *Pol. J. Environ. Stud.* 24 (2015) 1379–1386.
- [13] M.V. Balarama Krishna, K. Chandrasekaran, G. Venkateswarlu, D. Karunasagar, A cost effective and rapid microwave-assisted acid extraction method for the multi-elemental analysis of sediments by ICP-AES and ICP-MS, *Analytical Methods.* 4 (10) (2012) 3290, <https://doi.org/10.1039/c2ay25393c>.
- [14] K. Richa, K. Smriti, K. Monika, P. Jaya, Physico-chemical analysis of Ganga River water, *Asian J. Biochem. Pharmaceutical Res.* 1 (2011) 232–239.
- [15] A. Walkley, I.A. Black, An examination of Degtjareff method for determining soil organic matter and a proposed modification of the chromic acid titration method, *Soil Sci.* 37 (1934) 29–37.
- [16] B.V. Subbiah, G.L. Asija, A rapid procedure for the estimation of available nitrogen in soil, *Curr. Sci.* 25 (1956) 259.
- [17] M.G. Goydaragh, R. Taghizadeh-Mehrjardi, A.A. Jafarzadeh, J. Triantafyllis, M. Lado, Using environmental variables and Fourier Transform Infrared Spectroscopy to predict soil organic carbon, *Catena* 202 (2021), 105280.
- [18] P.K. Shukla, R.R. Mishra, Influence of herbicides on microbial population and enzyme activities in potato (*Solanum tuberosum*) field soil, *J. Agric. Sci.* 67 (1997) 610–611.
- [19] P.K. Shetty, S.P. Magu, Effect of metalaxyl on soil microbial population, *J. Trop. Agric.* 38 (2000) 63–65.
- [20] H.H. Azzaz, H.A. Murad, A.M. Kholif, M.A. Hanfy, M.H.A. Gawad, Optimization of culture conditions affecting fungal cellulase production, *Research J. of Microbiol.* 7 (1) (2012) 23–31.
- [21] P.N. Bhattacharyya, D.K. Jha, Plant growth-promoting rhizobacteria (PGPR): emergence in agriculture, *World J. Microbiol. Biotechnol.* 28 (4) (2012) 1327–1350.
- [22] K. Balasubramanian, S. Sankaran, Effect of pendimethalin on soil microorganisms, *Indian Agric.* 45 (2001) 93–98.
- [23] J.N. Kumar, H. Soni, R.N. Kumar, I. Bhatt, Macrophytes in phytoremediation of heavy metal contaminated water and sediments in Parijej Community Reserve, Gujarat, India, *Turkish Journal of Fisheries and Aquatic Sciences* 8 (2) (2008).
- [24] N. Snežana, K. Ljiljana, N. Miloš, M. Miško, M. Smilja, P. Vladimir, Physico-chemical soil analysis of Rudovci region, *Geonauka.* 2 (2013) 22–27.
- [25] P.L.A. Satyavathi, M. Suryanarayan Reddy, Soil-site suitability for six major crops in Telangana Region of Andhra Pradesh, *J. Indian Soc. Soil Sci.* 52 (2004) 220–225.
- [26] M. Shamsudheen, G.S. Dasog, N.B. Tejaswini, Land use/ Land cover mapping in the coastal area of North Karnataka using Remote sensing data, *Journal of the Indian Society of Remote Sensing* 33 (2) (2005) 253–257.
- [27] S. Verma, P. Van der Zaag, Virtual water trade and India's water future 2050. Paper presented at the EGU General Assembly, session "Water resources assessment with special focus on developing countries". Vienna, 18 April. *Geophysical Research Abstracts* Vol. 9, No. 05601, 2007.
- [28] C.C. Tsui, C.F. Hsieh, Z.S. Chen, Relationship between Soil Properties and Landscape in a Lowland Rain Forest of Southern Taiwan, *Geoderma* 123 (2004) 131–142.
- [29] A. Louhi, A. Hammadi, M. Achouri, Determination of some heavy metal pollutants in sediments of the Seybouse River in Annaba, Algeria, *Air, Soil Water Res.* 5 (2012). ASWR-S10081.
- [30] H. Meghraj, P. Daneshwar, Isolation and Identification of Heavy Metals Tolerant Bacteria from Industrial and Agricultural Areas in Mauritius, *Curr. Res. Microbiol. Biotechnol.* 1 (2013) 119–123.
- [31] L. Ranjard, S. Nazaret, F. Gourbière, J. Thioulouse, P. Linet, A. Richaume, A soil microscale study to reveal the heterogeneity of Hg(II) impact on indigenous bacteria by quantification of adapted phenotypes and analysis of DNA community fingerprints FEMS, *Microbiol. Ecol.* 31 (2000) 107–115.
- [32] R.-A. Sandaa, V. Torsvik, Ø. Enger, Influence of long-term heavy-metal contamination on microbial communities in soil, *Soil Biol. Biochem.* 33 (3) (2001) 287–295.
- [33] E. Kandeler, D. Tschirko, K.D. Bruce, M. Stemmer, P.J. Hobbs, R.D. Bardgett, W. Amelung, Structure and function of the soil microbial community in microhabitats of a heavy metal polluted soil, *Biol. Fertil. Soils.* 32 (5) (2000) 390–400.
- [34] J.M. Becker, T. Parkin, C.H. Nakatsu, J.D. Wilbur, A. Konopka, Bacterial activity, community structure, and centimetre-scale spatial heterogeneity in contaminated soil, *Microb. Ecol.* 51 (2006) 220–231.
- [35] L. Ranjard, L. Lignier, R. Chaussod, Cumulative effect of short-term polymetal contamination on soil bacterial community structure, *Appl. Environ. Microbiol.* 72 (2006) 1684–1687.
- [36] L.A. Ogilvie, A. Grant, Linking pollution induced community tolerance (PICT) and microbial community structure in chronically metal polluted estuarine sediments, *Mar. Environ. Res.* 65 (2) (2008) 187–198.
- [37] K. Sajjaphan, K. Sajjaphan, M.J. Sadowsky, Structure and diversity of arsenic-resistant bacteria in an old tin mine area of Thailand, *J. Microbiol. Biotechnol.* 20 (1) (2010) 169–178.
- [38] S. Khan, A.-E.-L. Hesham, M. Qiao, S. Rehman, J.Z. He, Effect of Cd and Pb on soil microbial community structure and activities, *Environ. Sci. Polut. Res.* 17 (2010) 288–296.
- [39] E. Bååth, K. Arnebrant, Growth rate and response of bacterial communities to pH in limed and ash treated forest soils, *Soil Biol. Biochem.* 26 (8) (1994) 995–1001.
- [40] E. Bååth, Measurement of heavy metal tolerance of soil bacteria using thymidine incorporation into bacteria extracted after homogenization-centrifugation, *Soil Biol. Biochem.* 24 (11) (1992) 1167–1172.
- [41] P.K. Joshi, A. Swarup, S. Maheshwari, R. Kumar, N. Singh, Bioremediation of heavy metals in liquid media through fungi isolated from contaminated sources, *Indian J. Microbiol.* 51 (4) (2011) 482–487.
- [42] R. Carrillo-González, M.D.C.A. González-Chávez, Tolerance to and accumulation of cadmium by the mycelium of the fungi *Sclerotinia citrinum* and *Pisolithus tinctorius*, *Biol. Trace Elem. Res.* 146 (3) (2012) 388–395.
- [43] L. Chai, S. Huang, Z. Yang, B. Peng, Y. Huang, Y. Chen, Cr (VI) remediation by indigenous bacteria in soils contaminated by chromium-containing slag, *J. Hazard. Mater.* 167 (1–3) (2009) 516–522.
- [44] W. Ren, L.I. Peijun, Y. Geng, L.I. Xiaojun, Biological leaching of heavy metal from a contaminated soil by *Aspergillus niger*, *J. Hazard. Mater.* 167 (2009) 164–169.
- [45] Y. Wang, J. Shi, H. Wang, Q. Lin, X. Chen, Y. Chen, The influence of soil heavy metals pollution on soil microbial biomass, enzyme activity, and community composition near a copper smelter, *Ecotox. Environ. Safe.* 67 (1) (2007) 75–81.
- [46] G. Haferburg, E. Kothe, Microbes and metals: interactions in the environment, *J. Basic Microbiol.* 47 (6) (2007) 453–467.
- [47] V.N. Kavamura, E. Esposito, Biotechnological strategies applied to the decontamination of soils polluted with heavy metals, *Biotechnology Advances.* 28 (1) (2010) 61–69.
- [48] J. Solecka, J. Zajko, M. Postek, A. Rajnisz, Biologically active secondary metabolites from Actinomycetes, *Central European J. Biol.* 7 (2012) 373–390.
- [49] V.L. Colin, L.B. Villegas, C.M. Abate, Indigenous microorganisms as potential bioremediators for environments contaminated with heavy metals, *Int. Biodegradation Biodegradation.* 69 (2012) 28–37.
- [50] J. Hamed, M. Dehghani, F. Mohammadian, Isolation of Extremely Heavy Metal Resistant Strains of Rare Actinomycetes from High Metal Content Soils in Iran, *Int. J. Environ. Res.* 9 (2015) 475–480.
- [51] M.A. Polti, M.J. Amoroso, C.M. Abate, Chromium (VI) resistance and removal by actinomycete strains isolated from sediments, *Chemosphere.* 67 (4) (2007) 660–667.
- [52] A. Schmidt, G. Haferburg, M. Sineriz, D. Merten, G. Büchel, E. Kothe, Heavy metal resistance mechanisms in actinobacteria for survival in AMD contaminated soils", *Chemie der Erde—, Geochemistry.* 65 (2005) 131–144.
- [53] N.W. So, J.Y. Rho, S.Y. Lee, I.C. Hancock, J.H. Kim, A lead-absorbing protein with superoxide dismutase activity from *Streptomyces subtritus*, *FEMS Microbiol. Lett.* 194 (2001) 93–98.
- [54] V.H. Albarracín, B. Winik, E. Kothe, M.J. Amoroso, C.M. Abate, Copper bioaccumulation by the actinobacterium *Amycolatopsis* sp. AB0, *J. Basic Microbiol.* 48 (5) (2008) 323–330.

- [55] V.H. Albarracín, P. Alonso-Vega, M.E. Trujillo, M.J. Amoroso, C.M. Abate, *Amycolatopsis tucumanensis* sp. nov., a copper-resistant actinobacterium isolated from polluted sediments, *Int. J. Systematic Evol. Microbiol.* 60 (2010) 397–401.
- [56] J.S. Dávila Costa, V.H. Albarracín, C.M. Abate, Responses of environmental *Amycolatopsis* strains to copper stress, *Ecotoxicol. Environ. Safety.* 74 (7) (2011) 2020–2028.
- [57] M.S. Fuentes, C.S. Benimeli, S.A. Cuozzo, M.J. Amoroso, Isolation of pesticide degrading actinomycetes from a contaminated site: bacterial growth, removal and dechlorination of organochlorine pesticides, *Int. Biodeterioration Biodegradation.* 64 (6) (2010) 434–441.